IDTA-02062

Interface Connector 1.0

October 2026

SPECIFICATION

Submodel Template

IDTA approved

- 100% AAS compliant
- Consistent & interoperable
- Released by the AAS experts

Submodel Template of the
Asset Administration Shell

IDTA-02062-1 V1.0

Imprint

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Chapter 1. General

1.1. About this document

This document is a part of an overall specification series [4]. Each part specifies the contents of a Submodel Template (SMT). The specifications of the Asset Administration Shell (AAS) are the basis for the Submodel Template specifications, see [3].

The target audience of the specification are developers and editors of technical documentation and manufacturer information, which are describing assets by means of the Asset Administration Shell (AAS) and therefore need to create a Submodel instance with a hierarchy of SubmodelElements. This document especially details on the question, which SubmodelElements with which semantic identification shall be used for this purpose.

This SMT will only be fully supported as of metamodel V3.1. V3.1 allows to assign idShorts to Elements within a SubmodelElementList (SML).

1.2. Scope of the Submodel

This Submodel template aims at interoperable provision of product connections points between manufacturers and users of industrial product types and items, particularly industrial components. These industrial product types and items are typically provided by manufacturers and suppliers, including dealers, and are used by industrial users, e.g. original equipment manufacturers (OEMs), system integrators and producing enterprises (industrial end users). The product types are typically used to provide more than one product instance, however special cases such as configuration are applicable. Product types have individual types and numbers of connection points, which are important to know for industrial users especially during engineering.

The aim of this Submodel is to digitialize and interoperably convey sets of minimal required information and to ease the handover of these information sets, to make it possible to efficiently filter, monitor and store this information in software systems on industrial user’s side. The intended use-case is, that a manufacturer of industrial product types and items provides data for all connection points related to the product available in a way, that these are interoperable and unambiguously understood by the other market participants, such as OEMs, system integrators or operators of industrial equipment. This Submodel template specifies a basic set of SubmodelElements in order to bring about the necessary information according to this use-case.

1.3. Explanations on used UML diagrams

For clarity and an improved legibility readers suggested to go through this section at first before reading the following chapters.

UML diagrams feature box-like elements, called "classes". These classes, typically Submodels, SubmodelElementCollections or SubmodelElementLists, typically feature a set of Properties or further SubmodelElements. These elements can have specific cardinalities.

The single classes are hierarchally organized by aggregation relations, these can be seen as "contains" relation.

For a further overview on UML diagrams please refer to [6] and [10].

Further details about used table formats please refer to Annex A.

Chapter 2. Information set for Submodel ”InterfaceConnector”

2.1. General

The following subsections introduce the basic ideas behind the SM InterfaceConnector, explain the scope of the connectors considered, and provide an example.

2.1.1. Intention of the SMT

The objective of the submodel is to specify the physical interfaces of components. The components considered include, for example, automation components such as I/O devices as well as cables. For each component, in particular the following aspects are to be described:

- Number of possible connections
- Type of connection
- Technical characteristics of the connections
- Connection configurations

As part of defining the submodel, the intention was to standardize the structure of the interface description and, at the same time, to establish a comprehensive set of generally applicable properties. In addition, the submodel enables the definition of supplier-specific properties.

The submodel is designed to support various end-user use cases. These include, for example, populating PLM systems with master data for the components, providing information for engineering tools, and supporting the ordering process.

Figure 1 gives an overview about the intended use of the submodel.

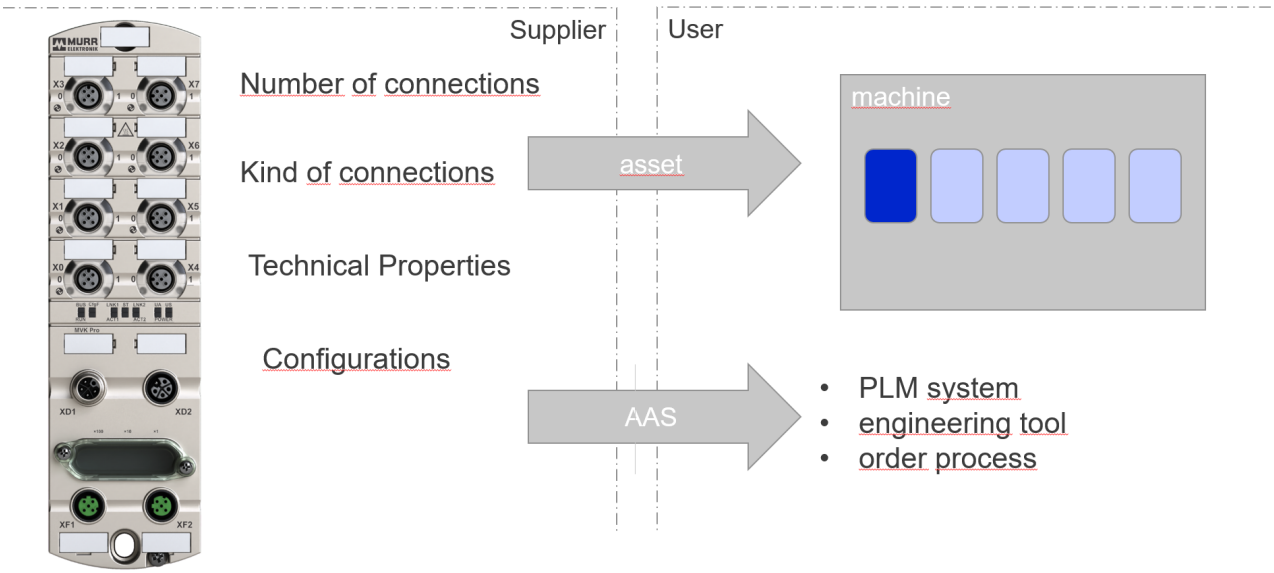


Figure 1. Overview Intention of the SMT

At a fundamental level, the architecture of the SM `InterfaceConnectors` provides that each connector is described individually. Each connector is assigned SMC that each describe a physical aspect. These may include, for example, the mechanical or pneumaticc characteristics of the connector. In general, every connector should have a mechanical description. The corresponding properties are added within the respective SMs.

Furthermore, each connector should have a SMC containing the relevant connection points. Under this collection, all connection points are described along with their respective properties. Individual connection points can be grouped into different configurations. Each connector has exactly one active configuration.

A typical example of a connector description would be an M12 plug, which has both mechanical characteristics and electrical connection points. Within the “Mechanical Connection” submodel collection, all relevant information about the M12 connection would be defined. In addition, the corresponding electrical pins would be described as connection points.

Details on the predefined properties can be found in Section 3.

2.1.2. Considered Connection Types

The SMT `InterfaceConnectors` allows the description of different interface connectors. Specifically, it is possible to describe the following interface types and combinations of their respective properties.

- Electric connection
- Mechanic connection
- Pneumatic connection
- Hydraulic connection
- Optic connection

Within the description of the interfaces the main physical properties of them are specified.

2.1.3. Example use for the SMT

The following section explains how to use the SM Interface Connector using the example of a cable(head). This is described in SM as entity "Connection_0."

The cable shown in Figure 2 is an M12 connector with 4 pins.

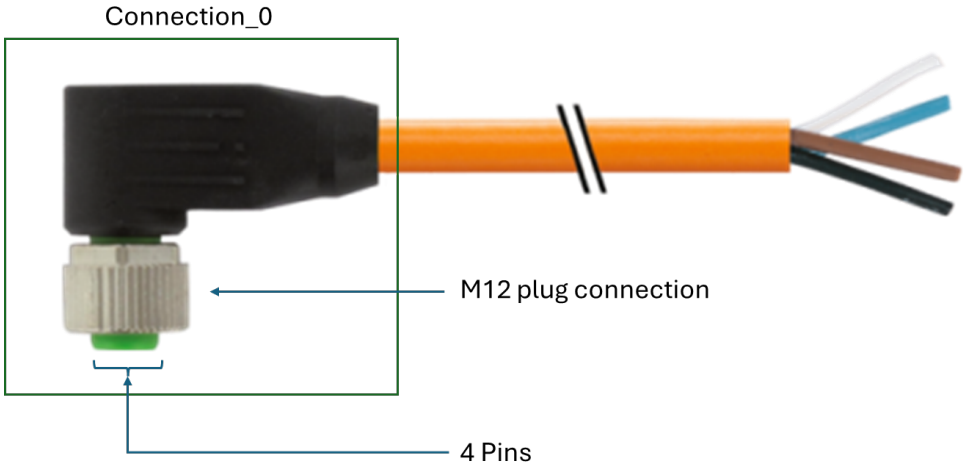


Figure 2. Example cable with M12 interface

In the Figure 3, the submodel `InterfaceConnectors` is shown using the example. This submodel describes the device’s interfaces/connectors. It contains a detailed description of "Connection_0".

Within "Connection_0", several sub-sections are defined:

- The submodel `Connector_related` with 7 elements: Information related to the connector (e.g., type, coding/keying, manufacturer data, standards, part numbers).

- The submodel `Mechanical_connection` with 2 elements: Mechanical aspects of the connection (e.g., locking mechanism, thread, mounting/installation dimensions).
- The submodel `ConnectionPoints` with 2 elements: The specific connection/contact points where electrical signals or power are present.

Overall, the representation explains how the digital metadata for the connections of a physical product are hierarchically organized within the AAS submodel `InterfaceConnectors`.

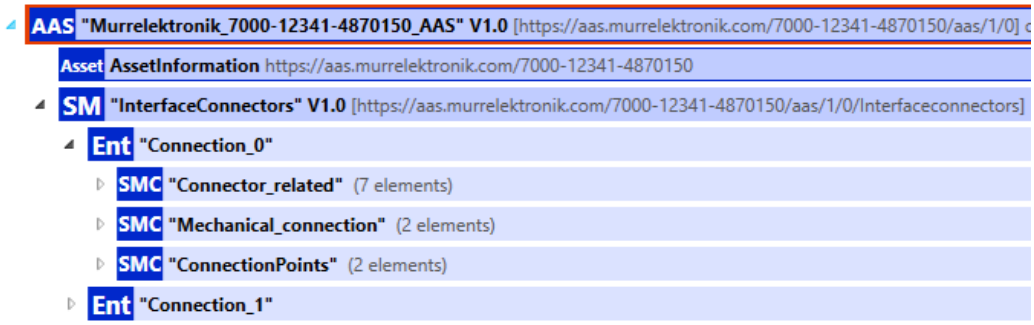


Figure 3. Example overview SM InterfaceConnectors

In the figure, Figure 4 the entity `Connection_0` is expanded, in particular the SMC `Connector_related` (7 elements). This section bundles the key master data and technical characteristics of the connector under consideration. Within `Connector_related`, the attributes are listed as properties and populated with concrete values:

- portLabel: Connector_0 — label/name of the connector.
- connectionType: M12 — connector type/design.
- numberOfConnectionPoints: 4 — number of contact/connection points (pins).
- cableOutlet: angled — cable outlet orientation (angled).
- coding: A — connector coding/keying.
- ipProtection: IP67, IP66K, IP65 — ingress protection ratings (dust/water protection).
- material: PUR — material specification (e.g., jacket/components).

Overall, the representation illustrates how connection information is structured hierarchically within the `InterfaceConnectors` submodel, and how the Connector_related` section provides the most important parameters of a specific connector using standardized properties.

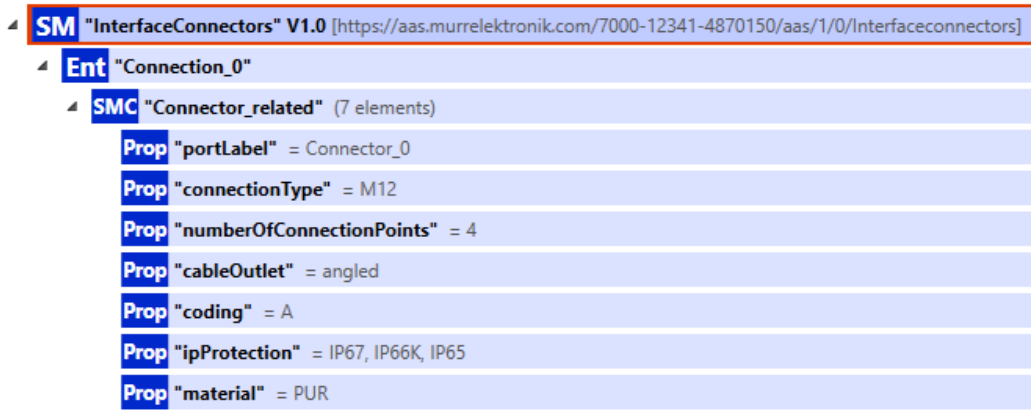


Figure 4. Example details SMC Connector_related

In the Figure 5, is the focus on the section `Mechanical_connection` (2 elements), which describes the mechanical way the connector is attached. Within `Mechanical_connection`, two groups are defined:

- `Screw_connection` (3 elements)
  - tighteningTorque: 0.6 N·m — recommended tightening torque for the screw coupling.
  - threadStandard: M12 — thread standard/size of the coupling.
  - typeOfFastening: SW13 — required tool size (wrench size 13).
- `Push-in_connection` (2 elements)
  - typeOfConnection: inserted — the connection is established by inserting the connector.
  - lockable: false — the push-in connection is not lockable.

Overall, the view illustrates how the `InterfaceConnectors` submodel structures mechanical connection metadata for a specific connector (`Connection_0`) in a clear, hierarchical way.

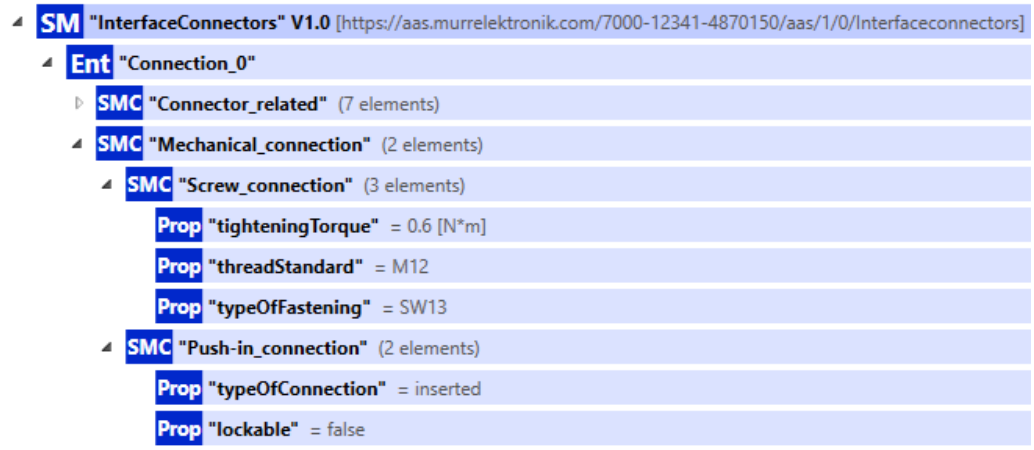


Figure 5. Example details SMC Mechanical_connection

In the Figure 6, the SMC ´ConnectionPoints´ (2 elements), which structures the connector's individual contact points (pins) and their configuration is depiced.

Within`ConnectionPoints`, a reference element ´used_configuration´ links to the configuration that is currently applied for this connection (i.e., which set of connection-point definitions is used for ´Connection_0).

The SMC `Configuration_0` (4 elements) contains the actual connection-point entities. The entities `ConnectionPoint_0` to `ConnectionPoint_3` represent the individual pins. In the expanded example `ConnectionPoint_0`, a property "label" is set to "Pin_1". I

Overall, the view illustrates how the `InterfaceConnectors` submodel represents the pin-level structure of a connector in a hierarchical and configurable way, including.

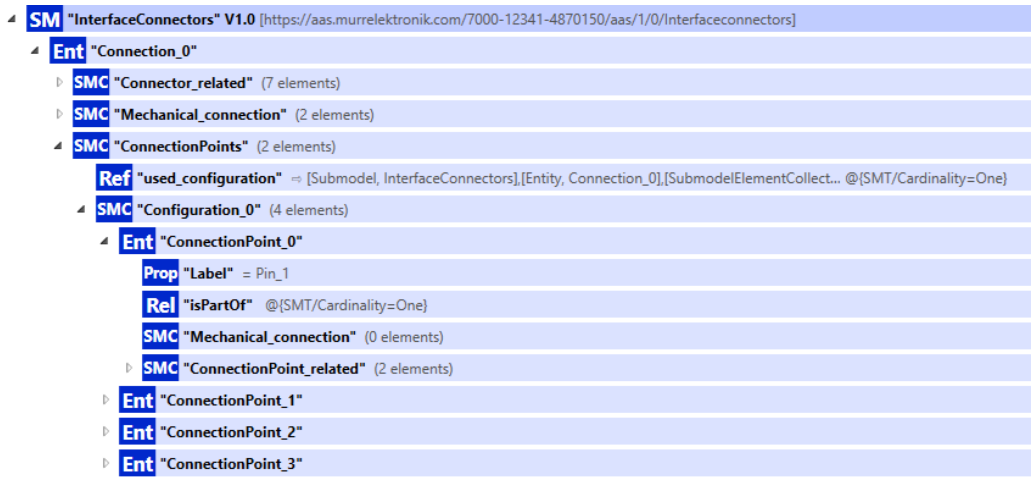


Figure 6. Example details SMC ConnectionPoints

2.2. Overview UML model

The SubmodelElements described in section 3 are structured in the following way (see Figure 7):

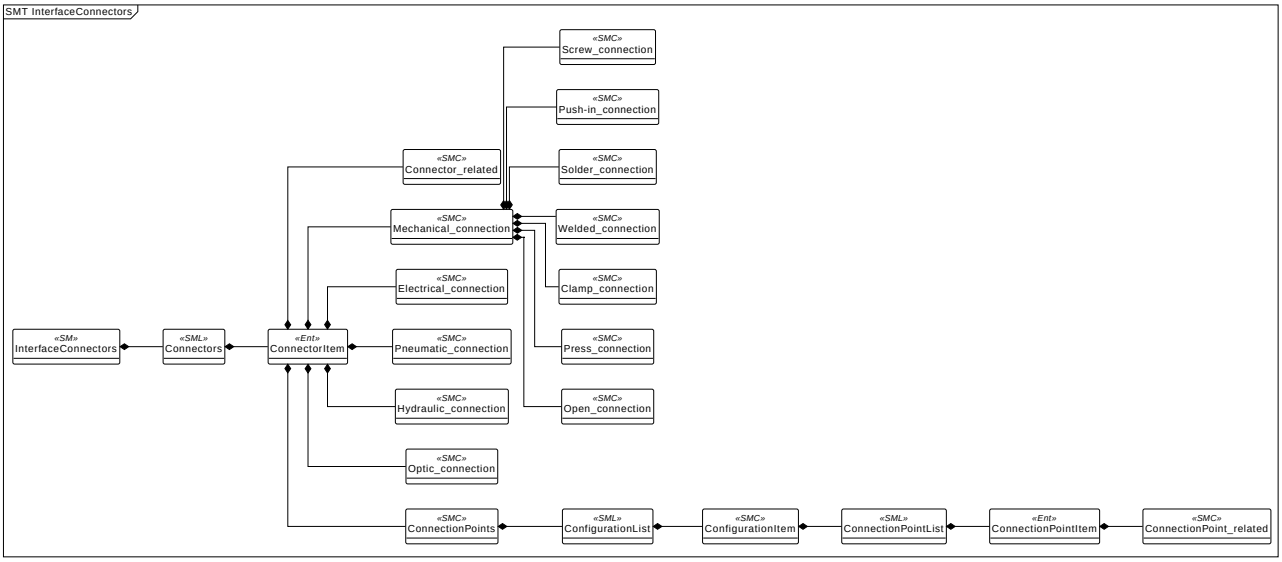


Figure 7. UML-Diagram for Submodel "Interface Connectors" for batteries

Chapter 3. Information structures and attributes

3.1. Structure of the Submodel "InterfaceConnector"

Figure 8 shows the UML-diagram for SMC `InterfaceConnector`.

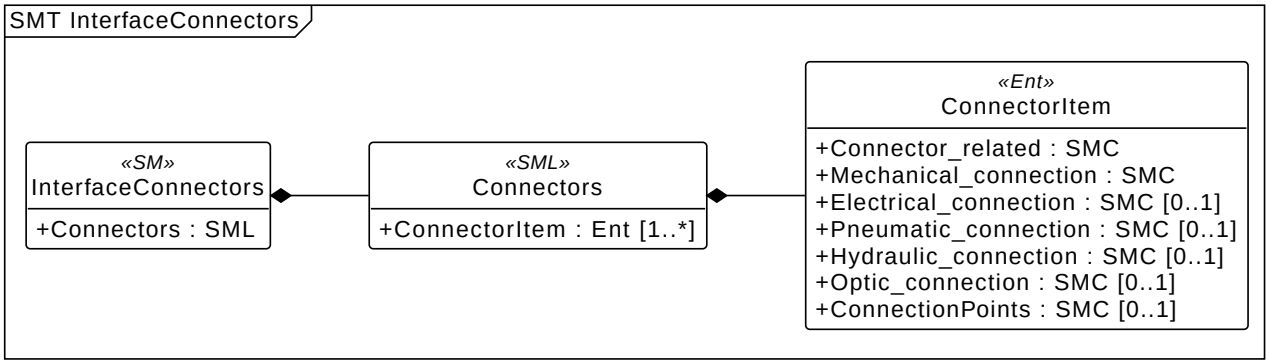


Figure 8. UML-Diagram for Submodel "InterfaceConnector"

In Table 1 the relevant properties and sub elements are defind which need to be set.

Table 1. Child Elements and Attributes for Submodel "InterfaceConnectors"

idShort:	**InterfaceConnectors**		
Class:	Submodel		
semanticId:	https://admin-shell.io/idta/cds/InterfaceConnectors/1		
Parent:	-		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SML] Connectors	https://admin-shell.io/idta/cds/Connectors/1 List of connectors described by the Interface Connectors submodel template.	[] 1 elements	1

Table 2 shows the elements of the SubmodelElementList `Connectors`.

Table 2. Child Elements and Attributes for SML "Connectors"

idShort:	**Connectors**
Class:	SubmodelElementList
semanticId:	https://admin-shell.io/idta/cds/Connectors/1
Parent:	InterfaceConnectors

Explanation:			
Element details:	typeValueListElement=Entity		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Ent] ConnectorItem	https://admin-shell.io/idta/cds/Connector/1 Component or interface used to establish a mechanical, electrical, pneumatic, hydraulic, or optical connection between compatible counterparts.	[] 7 elements	1..*

3.2. Structure of the Entity "ConnectorItem"

Figure 9 shows the UML-diagram for Ent `ConnectorItem`.

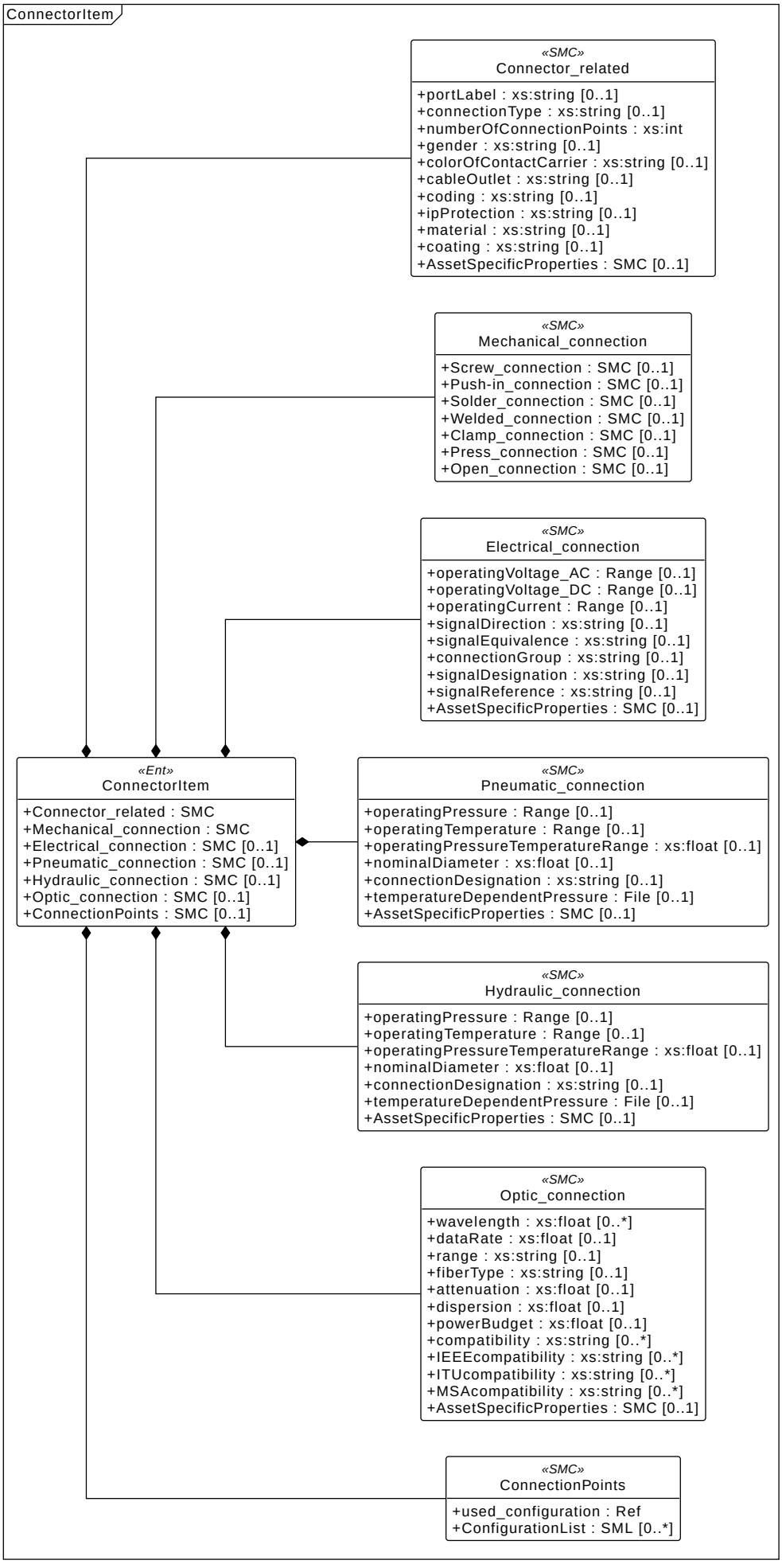


Figure 9. UML-Diagram for Entity "ConnectorItem"

In Table 3 the relevant properties and sub elements are defind which need to be set.

Table 3. Child Elements and Attributes for Entity "ConnectorItem"

idShort:	**ConnectorItem**		
Class:	Entity		
semanticId:	https://admin-shell.io/idta/cds/Connector/1		
Parent:	Connectors		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] Connector_related	https://admin-shell.io/idta/cds/Connector_related/1 The submodel collection shall be used to specify all general properties of the connector.	[] 11 elements	1
[SMC] Mechanical_connection	https://admin-shell.io/idta/cds/Mechanical/1 The submodel collection shall be used to specify a mechanical connection (or connection point).	[] 7 elements	1
[SMC] Electrical_connection	https://admin-shell.io/idta/cds/Electrical/1 The submodel collection shall be used to specify an electrical connection (or connection point).	[] 9 elements	0..1
[SMC] Pneumatic_connection	https://admin-shell.io/idta/cds/Pneumatic/1 The submodel collection shall be used to specify a pneumatic connection (or connection point).	[] 7 elements	0..1
[SMC] Hydraulic_connection	https://admin-shell.io/idta/cds/Hydraulic/1 The submodel collection shall be used to specify a hydraulic connection.	[] 7 elements	0..1
[SMC] Optic_connection	https://admin-shell.io/idta/cds/OpticConnection/1 The submodel collection shall be used to specify a optic connection.	[] 12 elements	0..1
[SMC] ConnectionPoints	https://admin-shell.io/idta/cds/ConnectionPoints/1 The submodel collection shall be used to specify all connection points that belong to a connection.	[] 2 elements	0..1

3.3. Attributes of the SMC Connector_related

Figure 10 shows the UML-diagram for SMC `Connector_related`.

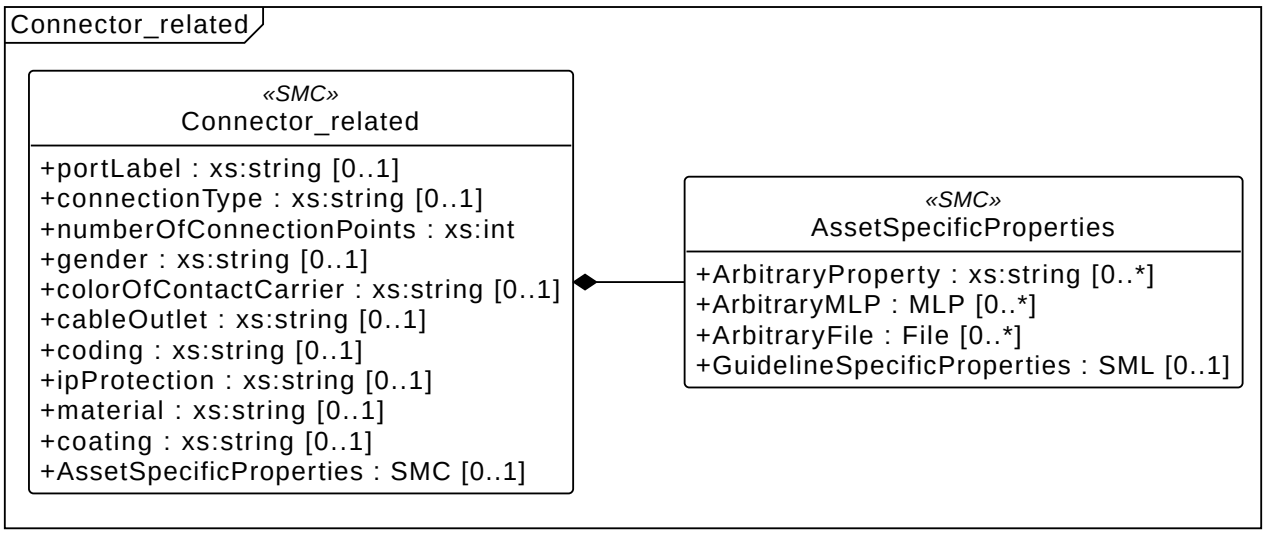


Figure 10. UML-Diagram for SMC Connector_related

In Table 4 the relevant properties and sub elements are defind which need to be set.

Table 4. Child Elements and Attributes for SMC "Connector_related"

idShort:	**Connector_related**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/Connector_related/1		
Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] portLabel	https://admin-shell.io/idta/cds/portLabel/1 This property defines the label of the connector. Example values are "X1", "XD1" or "XF1"	[String]	0..1
[Prop] connectionType	https://admin-shell.io/idta/cds/connectionType/1 This property defines the type of the connector at the port. If the connector is a standardized connection e.g. "M12 A-coded" this shall be the value of the property.	[String]	0..1
[Prop] numberOfConnectionPoints	https://admin-shell.io/idta/cds/numberOfConnectionPoints/1 This property defines the number of connection points at the connector. Number of physical or logical connection points provided by the connector.	[Int]	1

[Prop] gender	https://admin-shell.io/idta/cds/gender/1 This property specifies the gender of the port. Permitted values are "male" and "female".	[String]	0..1
[Prop] colorOfContact Carrier	https://admin-shell.io/idta/cds/colorOfContactCarrier/1 This property specifies the color of the contact carrier. Example values are "red" or "white"	[String]	0..1
[Prop] cableOutlet	https://admin-shell.io/idta/cds/cableOutlet/1 This property specifies the cable outlet. Specifies the direction or position at which the cable exits the connector.	[String]	0..1
[Prop] coding	https://admin-shell.io/idta/cds/coding/1 This property specifies coding of the connector. It should use coding information according to an international standard. Coding of the connector used to prevent incorrect mating with incompatible counterparts.	[String]	0..1
[Prop] ipProtection	https://admin-shell.io/idta/cds/ipProtection/1 This property specifies IP protection of the connector according to EN IEC 60529. Example value: IP67.	[String]	0..1
[Prop] material	https://admin-shell.io/idta/cds/material/1 This property specifies the material of the connector or connection point. Material of the described connector component.	[String]	0..1
[Prop] coating	https://admin-shell.io/idta/cds/coating/1 This property specifies the coating of the connector or connection point. Surface coating or plating applied to the described connector component.	[String]	0..1
[SMC] AssetSpecificP roperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.4. Attributes of the SMC "Mechanical_connection"

Figure 11 shows the UML-diagram for SMC `Mechanical_Connection`.

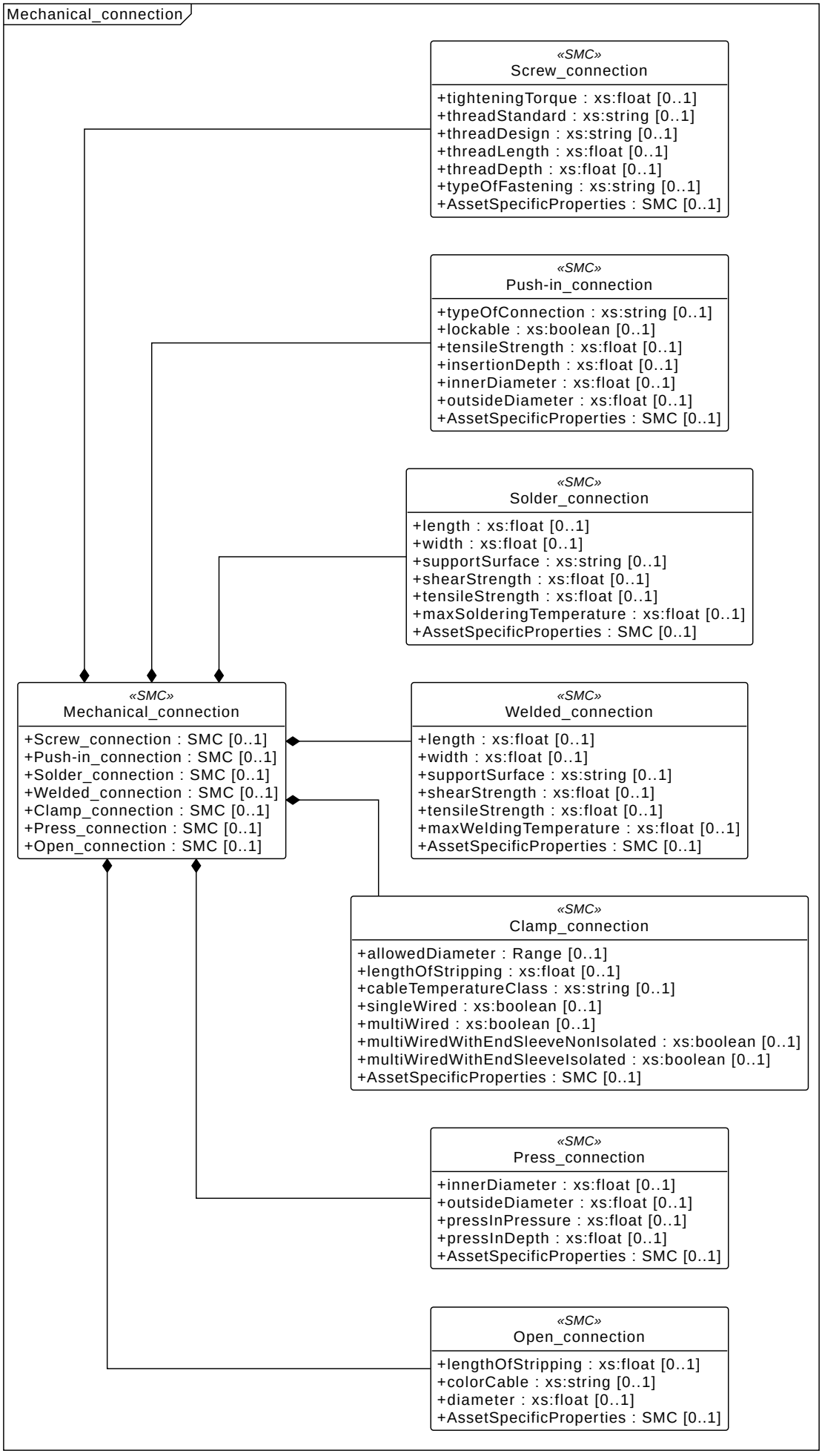


Figure 11. UML-Diagram for SMC "Mechanical_Connection"

In Table 5 to Table 12 the relevant properties and sub elements are defind which need to be set for the diffent types of mechanical connections.

Table 5. Child Elements and Attributes for SMC "Mechanical_connection"

idShort:	**Mechanical_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/Mechanical/1		
Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] Screw_connection	https://admin-shell.io/idta/cds/ScrewConnection/1 The submodel collection shall be used to specify a screw connection (or connection point).	[] 7 elements	0..1
[SMC] Push-in_connection	https://admin-shell.io/idta/cds/Push-inConnection/1 The submodel collection shall be used to specify a push-in connection (or connection point).	[] 7 elements	0..1
[SMC] Solder_connection	https://admin-shell.io/idta/cds/SolderConnection/1 The submodel collection shall be used to specify a solder connection (or connection point).	[] 7 elements	0..1
[SMC] Welded_connection	https://admin-shell.io/idta/cds/WeldedConnection/1 The submodel collection shall be used to specify a welded connection (or connection point).	[] 7 elements	0..1
[SMC] Clamp_connection	https://admin-shell.io/idta/cds/ClampConnection/1 The submodel collection shall be used to specify a clamp connection (or connection point).	[] 8 elements	0..1
[SMC] Press_connection	https://admin-shell.io/idta/cds/PressConnection/1 The submodel collection shall be used to specify a press connection (or connection point).	[] 5 elements	0..1
[SMC] Open_connection	https://admin-shell.io/idta/cds/OpenConnection/1 The submodel collection shall be used to specify an open connection (or connection point).	[] 4 elements	0..1

Table 6. Child Elements and Attributes for SMC "Screw_connection"

idShort:	**Screw_connection**
Class:	SubmodelElementCollection

semanticId:	https://admin-shell.io/idta/cds/ScrewConnection/1
Parent:	Mechanical_connection
Explanation:	
Element details:	-

[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] tighteningTorque	https://admin-shell.io/idta/cds/tighteningTorque/1 This property specifies tightening torque of the screw connection. The property shall be specified in N*m.	[Float]	0..1
[Prop] threadStandard	https://admin-shell.io/idta/cds/threadStandard/1 This property specifies the standard that is used for the screw connection., Example values are "R-Gewinde nach EN 10226-1" or "ISO 7/1"	[String]	0..1
[Prop] threadDesign	https://admin-shell.io/idta/cds/threadDesign/1 This property specifies the design of the screw connection, e.g. external thread or internal thread.	[String]	0..1
[Prop] threadLength	https://admin-shell.io/idta/cds/threadLength/1 This property specifies the thread length of an external thread. The property shall be specified in mm.	[Float]	0..1
[Prop] threadDepth	https://admin-shell.io/idta/cds/threadDepth/1 This property specifies thread depth of an internal thread. The property shall be specified in mm.	[Float]	0..1
[Prop] typeOfFastening	https://admin-shell.io/idta/cds/typeOfFastening/1 This property defines the type of fastening. Example values are "External socket" or "Hexagon socket"	[String]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

Table 7. Child Elements and Attributes for SMC "Push-in_connection"

idShort:	**Push-in_connection**

Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/Push-inConnection/1		
Parent:	Mechanical_connection		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] typeOfConnection	https://admin-shell.io/idta/cds/typeofConnection/1 This property defines the type of the push-in connection.	[String]	0..1
[Prop] lockable	https://admin-shell.io/idta/cds/lockable/1 This property defines if the push-in connection is lockable. true = connection is lockable; false = connection is not lockable	[Boolean]	0..1
[Prop] tensileStrength	https://admin-shell.io/idta/cds/tensileStrength/1 This property specifies the tensile strength for the push-in, solder or welded connection. The property shall be specified in N/mm^2.	[Float]	0..1
[Prop] insertionDepth	https://admin-shell.io/idta/cds/insertionDepth/1 This property specifies the insertion depth for the push-in connection. The property shall be specified in mm.	[Float]	0..1
[Prop] innerDiameter	https://admin-shell.io/idta/cds/innerDiameter/1 This property specifies the inner diameter for the push in or press connection. The property shall be specified in mm.	[Float]	0..1
[Prop] outsideDiameter	https://admin-shell.io/idta/cds/outsideDiameter/1 This property specifies the outside diameter for the push in or press connection. The property shall be specified in mm.	[Float]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

Table 8. Child Elements and Attributes for SMC "Solder_connection"

idShort:	**Solder_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/SolderConnection/1		
Parent:	Mechanical_connection		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] length	https://admin-shell.io/idta/cds/length/1 This property specifies the length of the solder or welded connection. The property shall be specified in mm.	[Float]	0..1
[Prop] width	https://admin-shell.io/idta/cds/width/1 This property specifies the width of the solder or welded connection. The property shall be specified in mm.	[Float]	0..1
[Prop] supportSurface	https://admin-shell.io/idta/cds/supportSurface/1 This property specifies the support surface for the solder or welded connection.	[String]	0..1
[Prop] shearStrength	https://admin-shell.io/idta/cds/shearStrength/1 This property specifies the shear strength of the solder or welded connection. The property shall be specified in N/mm^2.	[Float]	0..1
[Prop] tensileStrength	https://admin-shell.io/idta/cds/tensileStrength/1 This property specifies the tensile strength for the push-in, solder or welded connection. The property shall be specified in N/mm^2.	[Float]	0..1
[Prop] maxSolderingTemperature	https://admin-shell.io/idta/cds/maxSolderingTemperature/1 This property specifies the maximal temperature of the solder connection. The property shall be specified in °C.	[Float]	0..1

[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

Table 9. Child Elements and Attributes for SMC "Welded_connection"

idShort:	**Welded_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/WeldedConnection/1		
Parent:	Mechanical_connection		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] length	https://admin-shell.io/idta/cds/length/1 This property specifies the length of the solder or welded connection. The property shall be specified in mm.	[Float]	0..1
[Prop] width	https://admin-shell.io/idta/cds/width/1 This property specifies the width of the solder or welded connection. The property shall be specified in mm.	[Float]	0..1
[Prop] supportSurface	https://admin-shell.io/idta/cds/supportSurface/1 This property specifies the support surface for the solder or welded connection.	[String]	0..1
[Prop] shearStrength	https://admin-shell.io/idta/cds/shearStrength/1 This property specifies the shear strength of the solder or welded connection. The property shall be specified in N/mm^2.	[Float]	0..1
[Prop] tensileStrength	https://admin-shell.io/idta/cds/tensileStrength/1 This property specifies the tensile strength for the push-in, solder or welded connection. The property shall be specified in N/mm^2.	[Float]	0..1

[Prop] maxWeldingTemperature	https://admin-shell.io/idta/cds/maxWeldingTemperature/1 This property specifies the maximum temperature of the welded connection. The property shall be specified in °C.	[Float]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

Table 10. Child Elements and Attributes for SMC "Clamp_connection"

idShort:	**Clamp_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/ClampConnection/1		
Parent:	Mechanical_connection		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Range] allowedDiameter	https://admin-shell.io/idta/cds/allowedDiameter/1 This property specifies the permitted diameter range for the clamp connection. The property shall be specified in mm.	[Float] 0 ..	0..1
[Prop] lengthOfStripping	https://admin-shell.io/idta/cds/lengthOfStripping/1 This property specifies the stripping length of the cable. The property shall be specified in mm.	[Float]	0..1
[Prop] cableTemperatureClass	https://admin-shell.io/idta/cds/cableTemperatureClass/1 This property specifies the permitted cable temperature class for clamp connection. The property shall be specified in °C.	[String]	0..1
[Prop] singleWired	https://admin-shell.io/idta/cds/singleWired/1 This property specifies if the clamp connection is a single wired connection. true = single-wired false = not single-wired	[Boolean]	0..1

[Prop] multiWired	https://admin-shell.io/idta/cds/multiWired/1 This property specifies if the clamp connection is a multi wired connection. true = multi-wired false = not multi-wired	[Boolean]	0..1
[Prop] multiWiredWithEndSleeveNonIsolated	https://admin-shell.io/idta/cds/multiWiredWithEndSleeveNonIsolated/1 This property specifies if multi wire connection has a non-isolated end sleeve. true = is multi wired with end sleeve not isolated false = is not multi wired with end sleeve not isolated	[Boolean]	0..1
[Prop] multiWiredWithEndSleeveIsolated	https://admin-shell.io/idta/cds/multiWiredWithEndSleeveIsolated/1 This property specifies if multi wire connection has an isolated end sleeve. true = is multi wired with end sleeve isolated false = is not multi wired with end sleeve isolated	[Boolean]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

Table 11. Child Elements and Attributes for SMC "Press_connection"

idShort:	**Press_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/PressConnection/1		
Parent:	Mechanical_connection		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] innerDiameter	https://admin-shell.io/idta/cds/innerDiameter/1 This property specifies the inner diameter for the push in or press connection. The property shall be specified in mm.	[Float]	0..1

[Prop] outsideDiameter	https://admin-shell.io/idta/cds/outsideDiameter/1 This property specifies the outside diameter for the push in or press connection. The property shall be specified in mm.	[Float]	0..1
[Prop] pressInPressure	https://admin-shell.io/idta/cds/pressInPressure/1 Definition for pressInPressure The property shall be specified in N/m^2.	[Float]	0..1
[Prop] pressInDepth	https://admin-shell.io/idta/cds/pressInDepth/1 This property specifies the depth of the press connection. The property shall be specified in mm.	[Float]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

Table 12. Child Elements and Attributes for SMC "Open_connection"

idShort:	**Open_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/OpenConnection/1		
Parent:	Mechanical_connection		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] lengthOfStripping	https://admin-shell.io/idta/cds/lengthOfStripping/1 This property specifies the stripping length of the cable. The property shall be specified in mm.	[Float]	0..1
[Prop] colorCable	https://admin-shell.io/idta/cds/colorCable/1 This property specifies the cable color.	[String]	0..1
[Prop] diameter	https://admin-shell.io/idta/cds/diameter/1 This property specifies the diameter, The property shall be specified in mm.	[Float]	0..1

[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.5. Attributes of the SMC "Electrical_connection"

Figure 12 shows the UML-diagram for SMC `Electrical_connection`.

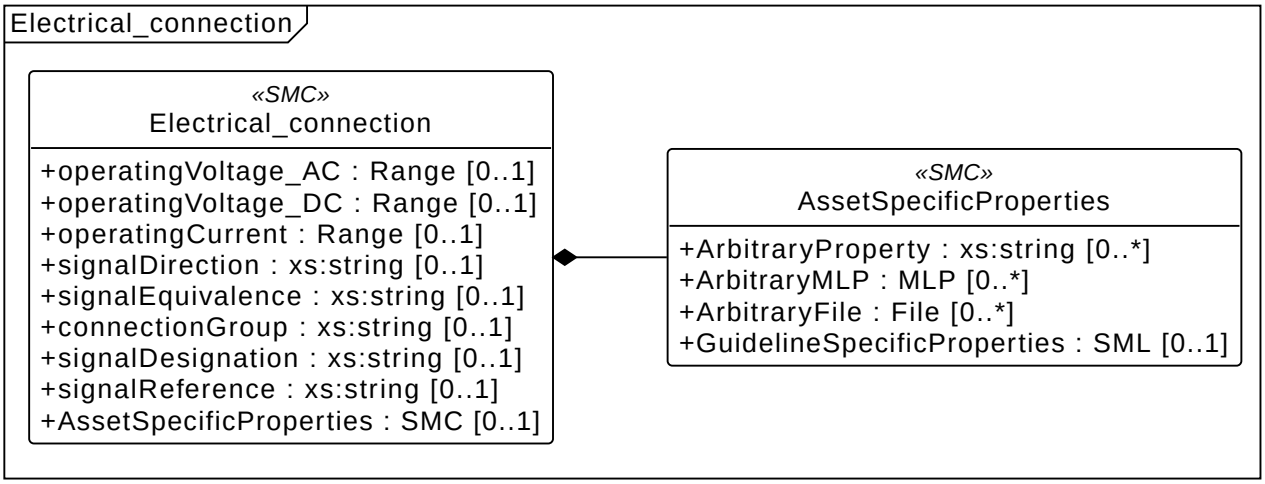


Figure 12. UML-Diagram for SMC "Electrical_connection"

In Table 13 the relevant properties and sub elements are defind which need to be set.

Table 13. Child Elements and Attributes for SMC "Electrical_connection"

idShort:	**Electrical_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/Electrical/1		
Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Range] operatingVoltage_AC	https://admin-shell.io/idta/cds/operatingVoltage_AC/1 This property specifies operating AC voltage of the electrical connection. The property shall be specified in V.	[Float] ..	0..1

[Range] operatingVoltage_DC	https://admin-shell.io/idta/cds/operatingVoltage_DC/1 This property specifies operating DC voltage of the electrical connection. The property shall be specified in V.	[Float] ..	0..1
[Range] operatingCurrent	https://admin-shell.io/idta/cds/operatingCurrent/1 This property specifies operating current of the electrical connection. The property shall be specified in A.	[Float] ..	0..1
[Prop] signalDirection	https://admin-shell.io/idta/cds/signalDirection/1 This property specifies signal direction of the electrical connection. Possible values include, for example, “In,” “Out,” and “IO.”	[String]	0..1
[Prop] signalEquivalence	https://admin-shell.io/idta/cds/signalEquivalence/1 This property specifies the signal equivalence—that is, the electrical potential that depends on the switch state—of the electrical connection. Identifier describing electrically equivalent signals or contacts.	[String]	0..1
[Prop] connectionGroup	https://admin-shell.io/idta/cds/connectionGroup/1 This property specifies the connection group to which the connector or connection point of the electrical connection belongs. Identifier of the group to which the connection or contact belongs.	[String]	0..1
[Prop] signalDesignation	https://admin-shell.io/idta/cds/signalDesignation/1 This property specifies the designation or identifier of the signal associated with the connector or connection point of the electrical connection. Designation assigned to the electrical signal.	[String]	0..1
[Prop] signalReference	https://admin-shell.io/idta/cds/signalReference/1 This property specifies the signal reference, i.e., the electrical potential of the electrical connection. Reference to the signal definition, signal source, or related documentation.	[String]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.6. Attributes of the SMC "Pneumatic_connection"

Figure 13 shows the UML-diagram for SMC `Pneumatic_connection`.

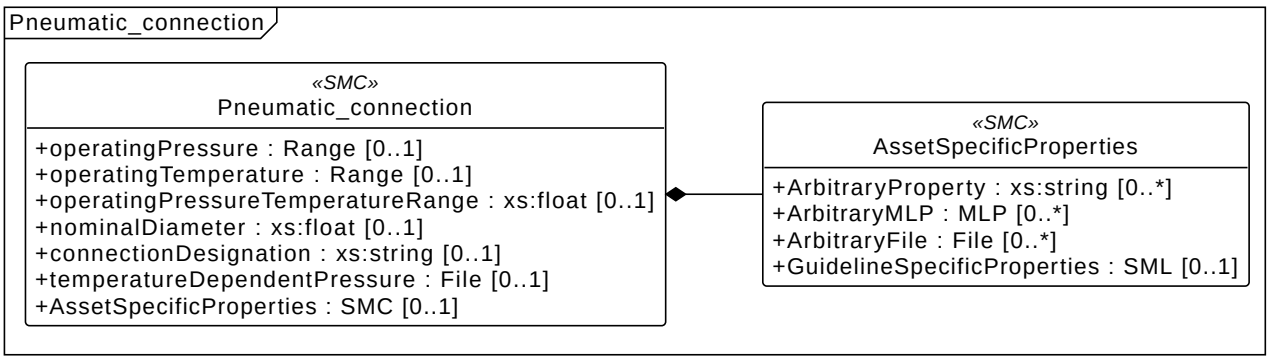


Figure 13. UML-Diagram for SMC "Pneumatic_Connection"

In Table 14 the relevant properties and sub elements are defind which need to be set.

Table 14. Child Elements and Attributes for SMC "Pneumatic_connection"

idShort:	**Pneumatic_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/Pneumatic/1		
Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Range] operatingPressure	https://admin-shell.io/idta/cds/operatingPressure/1 This property specifies operating pressure of the pneumatic or hydraulic connection. The property shall be specified in N/mm^2.	[Float] ..	0..1
[Range] operatingTemperature	https://admin-shell.io/idta/cds/operatingTemperature/1 This property specifies operating temperature of the pneumatic or hydraulic connection. The property shall be specified in °C.	[Float] ..	0..1
[Prop] operatingPressureTemperatureRange	https://admin-shell.io/idta/cds/operatingPressureTemperatureRange/1 This property specifies operating pressure of over the complete temperature range of the pneumatic or hydraulic connection. The property shall be specified in Bar.	[Float]	0..1

[Prop] nominalDiameter	https://admin-shell.io/idta/cds/nominalDiameter/1 This property specifies nominal diameter of the pneumatic or hydraulic connection. The property shall be specified in mm.	[Float]	0..1
[Prop] connectionDesignation	https://admin-shell.io/idta/cds/connectionDesignation/1 This property specifies the connection designation of the pneumatic or hydraulic connection.	[String]	0..1
[File] temperatureDependentPressure	https://admin-shell.io/idta/cds/temperatureDependentPressure/1 This file reference shall be used to reference a temperature pressure curve of the pneumatic or hydraulic connection. File containing temperature-dependent pressure information.	[]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.7. Attributes of the SMC "Hydraulic_connection"

Figure 14 shows the UML-diagram for SMC `Hydraulic_connection`.

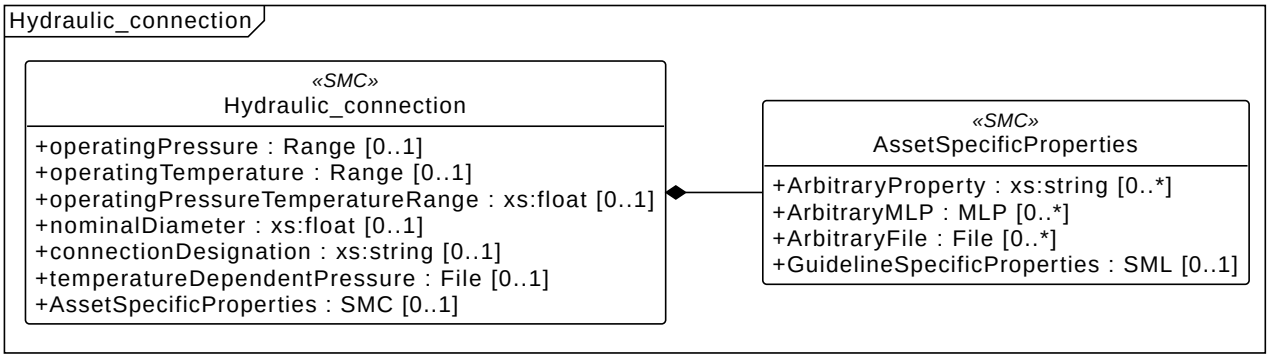


Figure 14. UML-Diagram for SMC "Hydraulic_Connection"

In Table 15 the relevant properties and sub elements are defind which need to be set.

Table 15. Child Elements and Attributes for SMC "Hydraulic_connection"

idShort:	**Hydraulic_connection**
Class:	SubmodelElementCollection
semanticId:	https://admin-shell.io/idta/cds/Hydraulic/1

Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Range] operatingPressure	https://admin-shell.io/idta/cds/operatingPressure/1 This property specifies operating pressure of the pneumatic or hydraulic connection. The property shall be specified in Bar.	[Float] ..	0..1
[Range] operatingTemperature	https://admin-shell.io/idta/cds/operatingTemperature/1 This property specifies operating temperature of the pneumatic or hydraulic connection. The property shall be specified in °C.	[Float] ..	0..1
[Prop] operatingPressureTemperatureRange	https://admin-shell.io/idta/cds/operatingPressureTemperatureRange/1 This property specifies operating pressure of over the complete temperature range of the pneumatic or hydraulic connection. The property shall be specified in Bar.	[Float]	0..1
[Prop] nominalDiameter	https://admin-shell.io/idta/cds/nominalDiameter/1 This property specifies nominal diameter of the pneumatic or hydraulic connection. The property shall be specified in mm.	[Float]	0..1
[Prop] connectionDesignation	https://admin-shell.io/idta/cds/connectionDesignation/1 This property specifies the connection designation of the pneumatic or hydraulic connection.	[String]	0..1
[File] temperatureDependentPressure	https://admin-shell.io/idta/cds/temperatureDependentPressure/1 This file reference shall be used to reference a temperature pressure curve of the pneumatic or hydraulic connection. File containing temperature-dependent pressure information.	[]	0..1

[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.8. Attributes of the SMC "Optic_connection"

Figure 15 shows the UML-diagram for SMC `Optic_connection`.

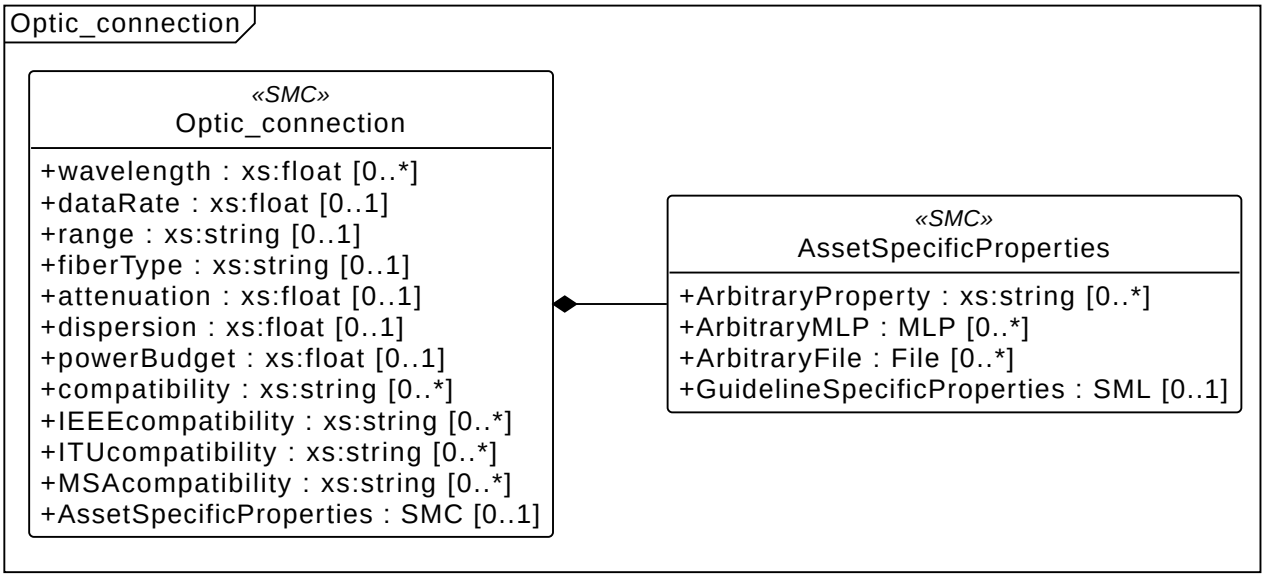


Figure 15. UML-Diagram for SMC "Optic_Connection"

In Table 16 the relevant properties and sub elements are defind which need to be set.

Table 16. Child Elements and Attributes for SMC "Optic_connection"

idShort:	**Optic_connection**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/OpticConnection/1		
Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.

[Prop] wavelength	https://admin-shell.io/idta/cds/wavelength/1 This property specifies wavelength of the optic connection. The property shall be specified in nm.	[Float]	0..*
[Prop] dataRate	https://admin-shell.io/idta/cds/dataRate/1 This property specifies data rate of the optic connection. The property shall be specified in Mbps.	[Float]	0..1
[Prop] range	https://admin-shell.io/idta/cds/range/1 Definition for range Maximum supported transmission range under the specified operating conditions.	[String]	0..1
[Prop] fiberType	https://admin-shell.io/idta/cds/fiberType/1 This property specifies fiber type of the optic connection. Type of optical fiber supported by the optical connection.	[String]	0..1
[Prop] attenuation	https://admin-shell.io/idta/cds/attenuation/1 This property specifies attenuation of the optic connection. The property shall be specified in dB.	[Float]	0..1
[Prop] dispersion	https://admin-shell.io/idta/cds/dispersion/1 This property specifies the dispersion of the optic connection. Optical dispersion characteristic of the supported fiber or optical connection. The property shall be specified in ps/(nm*km).	[Float]	0..1
[Prop] powerBudget	https://admin-shell.io/idta/cds/powerBudget/1 This property specifies power budget of the optic connection. Example value: 3.0 dB. The property shall be specified in dB.	[Float]	0..1
[Prop] compatibility	https://admin-shell.io/idta/cds/compatibility/1 Definition for compatibility of the optic connection to an internatiional standard. Compatibility to international standard.	[String]	0..*
[Prop] IEEEcompatibility	https://admin-shell.io/idta/cds/compatibility/IEEE/1 Definition for IEEE compatibility. The attribute specifies the name or number of an IEEE standard with which the optical link is compatible.	[String]	0..*
[Prop] ITUcompatibility	https://admin-shell.io/idta/cds/ITU/1 Definition for ITU compatibility. The attribute specifies the name or number of an ITU standard with which the optical link is compatible.	[String]	0..*

[Prop] MSAcompatibility	https://admin-shell.io/idta/cds/MSA/1 Definition for MSA compatibility. The attribute specifies the name or number of an MSA standard with which the optical link is compatible.	[String]	0..*
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.9. Attributes of the SMC "ConnectionPoints"

Figure 16 shows the UML-diagram for SMC `ConnectionPoints`.

Note: The SMC `ConnectionsPoints`collects all contact points of a connection. These can be grouped together in at least one or n configurations.

Note: The SMC `ConnectionsPoints`shall refrence to one configuration. This configuration should represent the current configuration of the connection.

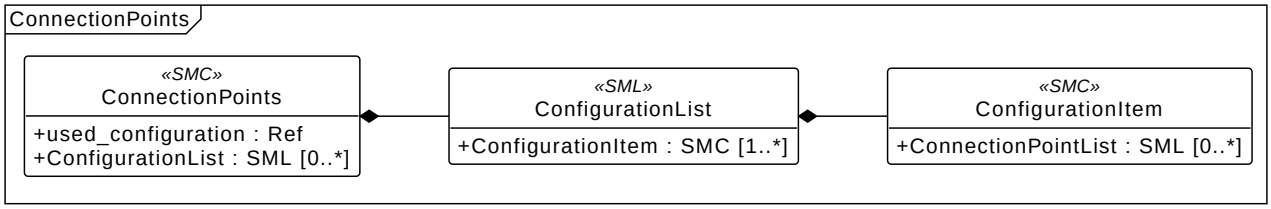


Figure 16. UML-Diagram for SMC "ConnectionPoints"

In Table 17 the relevant properties and sub elements are defind which need to be set.

Table 17. Child Elements and Attributes for SMC "ConnectionPoints"

idShort:	**ConnectionPoints**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/ConnectionPoints/1		
Parent:	ConnectorItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.

[Ref] used_configuration	https://admin-shell.io/idta/cds/used_configuration/1 The reference shall be used to reference the configuration of the interface connection. Reference to the default or used configuration.	[]	1
[SML] ConfigurationList	https://admin-shell.io/idta/cds/ConfigurationList/1 List of configurations associated with the connector. The submodel element list contains different configurations for the parent connector.	[] 1 elements	0..*

Table 18 shows the elements of the SubmodelElementList `ConfigurationList`.

Table 18. Child Elements and Attributes for SML "ConfigurationList"

idShort:	**ConfigurationList**		
Class:	SubmodelElementList		
semanticId:	https://admin-shell.io/idta/cds/ConfigurationList/1		
Parent:	ConnectionPoints		
Explanation:			
Element details:	typeValueListElement=SubmodelElementCollection		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] ConfigurationItem	https://admin-shell.io/idta/cds/Configuration/1 The submodel collection shall be used to specify a configuration of the interface connector.	[] 1 elements	1..*

Within the following sub sections the use of sub elements of the SMC `ConnectionPoint` is spechified in detail.

3.9.1. Attributes of the SMC "ConfigurationItem"

Figure 17 shows the UML-diagram for SMC `ConfigurationItem`.

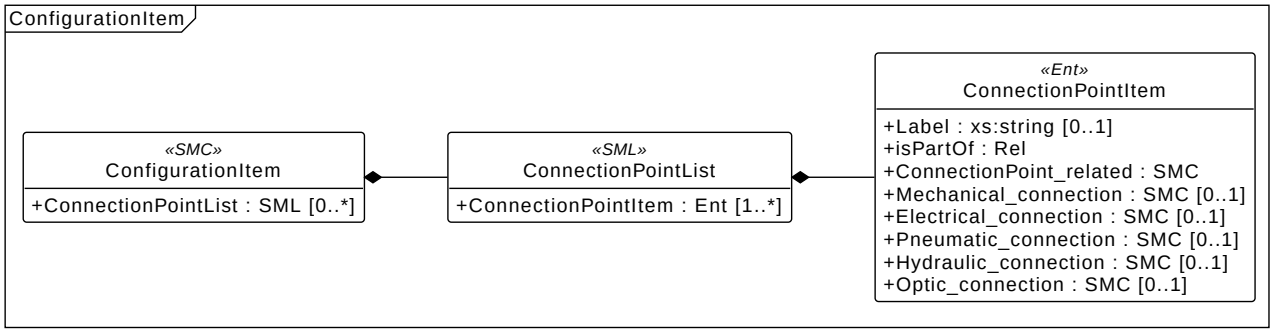


Figure 17. UML-Diagram for SMC "ConfigurationItem"

In Table 19 the relevant properties and sub elements are defind which need to be set.

Table 19. Child Elements and Attributes for SMC "ConfigurationItem"

idShort:	**ConfigurationItem**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/Configuration/1		
Parent:	ConfigurationList		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SML] ConnectionPointList	https://admin-shell.io/idta/cds/ConnectionPointList/1 List of connection points associated with a connector configuration. The submodel element list information regarding the different connection points of the parent connector.	[] 1 elements	0..*

Table 20 shows the elements of the SubmodelElementList `ConnectionPointList`.

Table 20. Child Elements and Attributes for SML "ConnectionPointList"

idShort:	**ConnectionPointList**		
Class:	SubmodelElementList		
semanticId:	https://admin-shell.io/idta/cds/ConnectionPointList/1		
Parent:	ConfigurationItem		
Explanation:			
Element details:	typeValueListElement=Entity		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Ent] ConnectionPointItem	https://admin-shell.io/idta/cds/ConnectionPoint/1 The Entity collection shall be used to specify a connection point.	[] 8 elements	1..*

3.9.2. Attributes of the Ent "ConnectionPointItem"

Figure 18 shows the UML-diagram for Ent `ConnectionPointItem`.

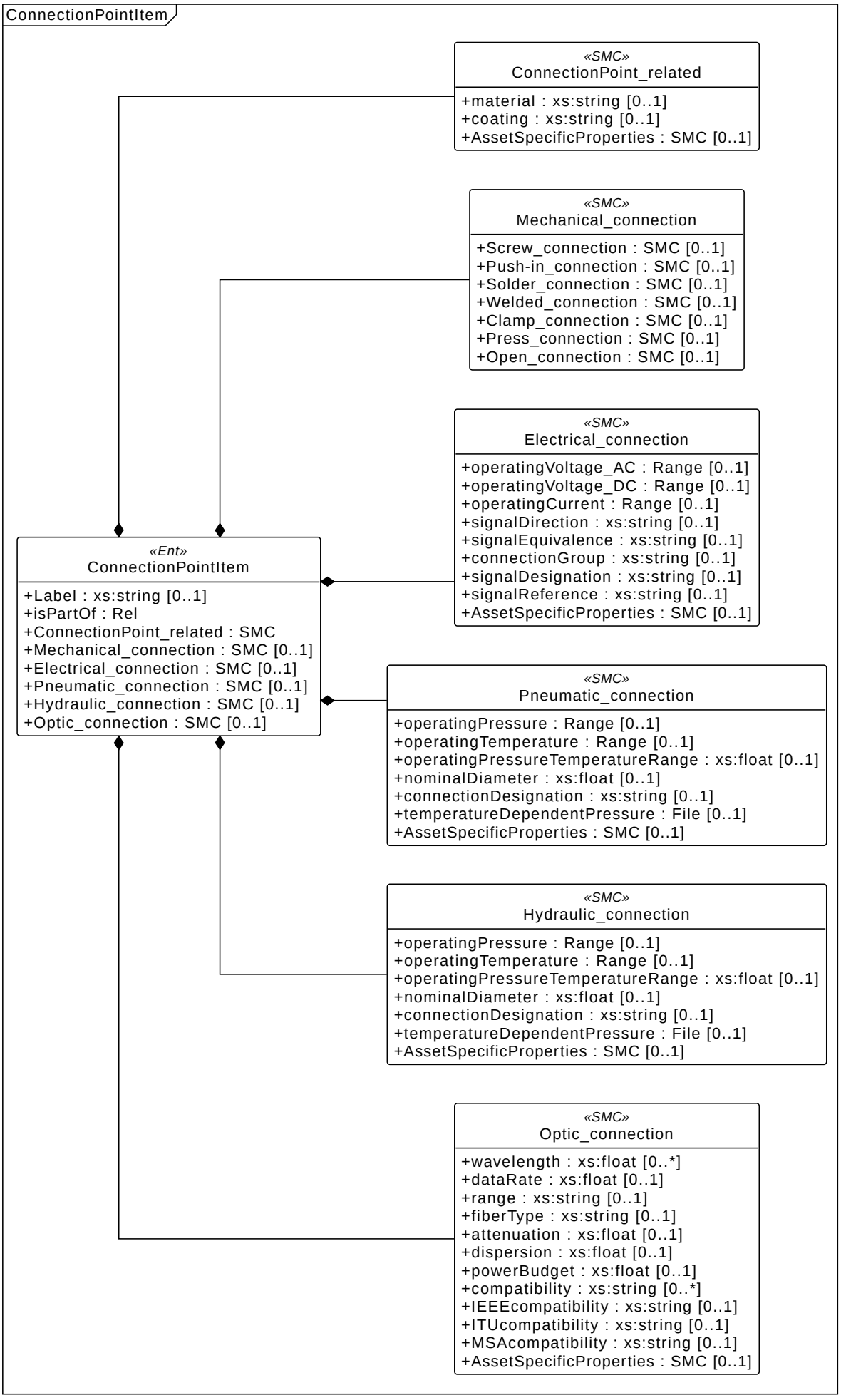


Figure 18. UML-Diagram for Entity "ConnectionPointItem"

In Table 21 the relevant properties and sub elements are defind which need to be set.

Table 21. Child Elements and Attributes for Entity "ConnectionPointItem"

idShort:	**ConnectionPointItem**		
Class:	Entity		
semanticId:	https://admin-shell.io/idta/cds/ConnectionPoint/1		
Parent:	ConnectionPointList		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] Label	https://admin-shell.io/idta/cds/Label/1 This property specifies the label of the connection point. Human-readable label assigned to the connection point.	[String] 1	0..1
[Rel] isPartOf	https://admin-shell.io/idta/cds/isPartOf/1 The relation shall be used to assign the connection point to an configuration. Relationship indicating that the source element is part of the target element.	[]	1
[SMC] ConnectionPoint_related	https://admin-shell.io/idta/cds/ConnectionPoint_related/1 The submodel collection shall be used to specify all general properties of the connection point.	[] 3 elements	1
[SMC] Mechanical_connection	https://admin-shell.io/idta/cds/Mechanical/1 The submodel collection shall be used to specify a mechanical connection (or connection point).	[] 7 elements	0..1
[SMC] Electrical_connection	https://admin-shell.io/idta/cds/Electrical/1 The submodel collection shall be used to specify an electrical connection (or connection point).	[] 9 elements	0..1
[SMC] Pneumatic_connection	https://admin-shell.io/idta/cds/Pneumatic/1 The submodel collection shall be used to specify a pneumatic connection (or connection point).	[] 7 elements	0..1
[SMC] Hydraulic_connection	https://admin-shell.io/idta/cds/Hydraulic/1 The submodel collection shall be used to specify a hydraulic connection.	[] 7 elements	0..1
[SMC] Optic_connection	https://admin-shell.io/idta/cds/OpticConnection/1 The submodel collection shall be used to specify a optic connection.	[] 12 elements	0..1

3.9.3. Attributes of the SMC "ConnectionPoint_related"

Figure 19 shows the UML-diagram for SMC `ConnectionPoint_related`.

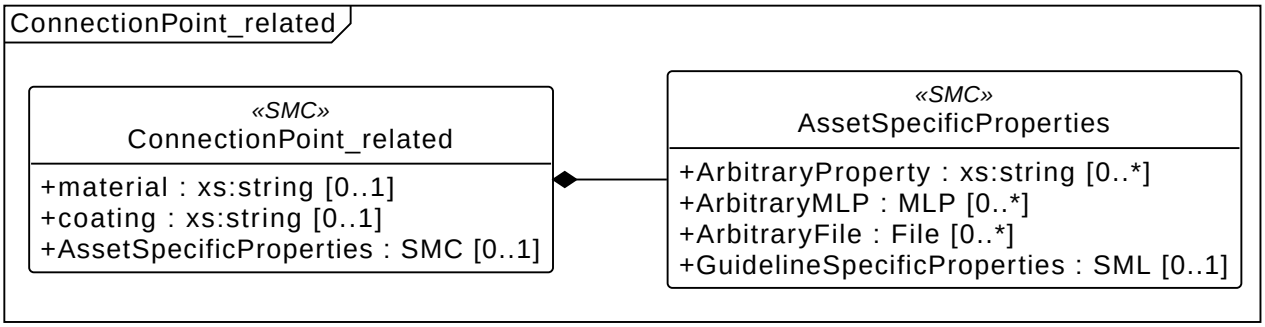


Figure 19. UML-Diagram for SMC "ConnectionPoint_related"

In Table 22 the relevant properties and sub elements are defind which need to be set.

Table 22. Child Elements and Attributes for SMC "ConnectionPoint_related"

idShort:	**ConnectionPoint_related**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/ConnectionPoint_related/1		
Parent:	ConnectionPointItem		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] material	https://admin-shell.io/idta/cds/material/1 This property specifies the material of the connector or connection point. Material of the described connector component.	[String]	0..1
[Prop] coating	https://admin-shell.io/idta/cds/coating/1 This property specifies the coating of the connector or connection point. Surface coating or plating applied to the described connector component.	[String]	0..1
[SMC] AssetSpecificProperties	https://admin-shell.io/idta/cds/AssetSpecificProperties/1 supplementalSemanticId: 0173-1#01-AGZ672#004, 0173-1#02-ABI218#003 Group of properties that are listed on the conection or connection point information and are grouped based on guidelines The SMC shall be used to define user / asset-specific properties.	[] 4 elements	0..1

3.10. Properties of the SMC “AssetSpecificProperties”

Figure 20 shows the UML-diagram defining the relevant properties which need to be set for asset specific properties.

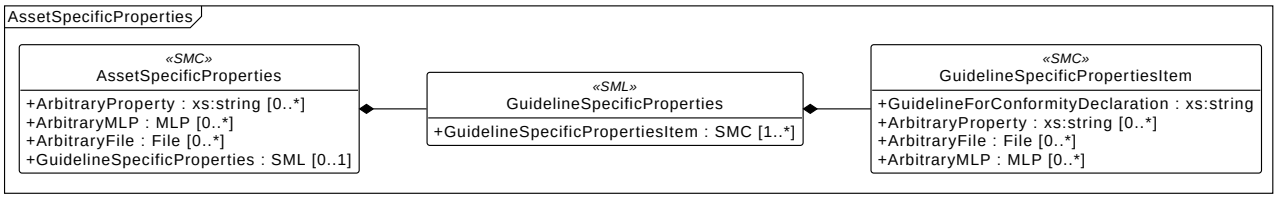


Figure 20. UML for asset specific properties and sub-structures

For the asset specific properties, the SubmodelElements are described as follows. The table convention is explained in Annex A.2.

Table 23. Child Elements and Attributes for SMC "AssetSpecificProperties"

idShort:	**AssetSpecificProperties**		
Class:	SubmodelElementCollection		
semanticId:	https://admin-shell.io/idta/cds/AssetSpecificProperties/1		
Parent:	Connector_related		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] ArbitraryProperty	https://admin-shell.io/SMT/General/ArbitraryProp Note: Every property can be used. Note: The idShort is arbitrary Note: The use of a displayName is recommended.	[String]	0..*
[MLP] ArbitraryMLP	https://admin-shell.io/SMT/General/ArbitraryMLP Note: Every multilingual property can be used. Note: The idShort is arbitrary Note: The use of a displayName is recommended.	[] ""sample""@en	0..*
[File] ArbitraryFile	https://admin-shell.io/SMT/General/ArbitraryFile Note: Every file can be used. The idShort is arbitrary Note: The use of a displayName is recommended.	[]	0..*

[SML] GuidelineSpecificProperties	https://admin-shell.io/idta/cds/GuidelineSpecificProperties/1 List of properties that are specific to an applicable guideline and are not covered by the standardized information model. The SMC shall be used to define guideline-specific properties.	[] 1 elements	0..1

3.11. Properties of the SMC “GuidelineSpecificProperties”

Figure 21 shows the UML-diagram defining the relevant properties which need to be set.

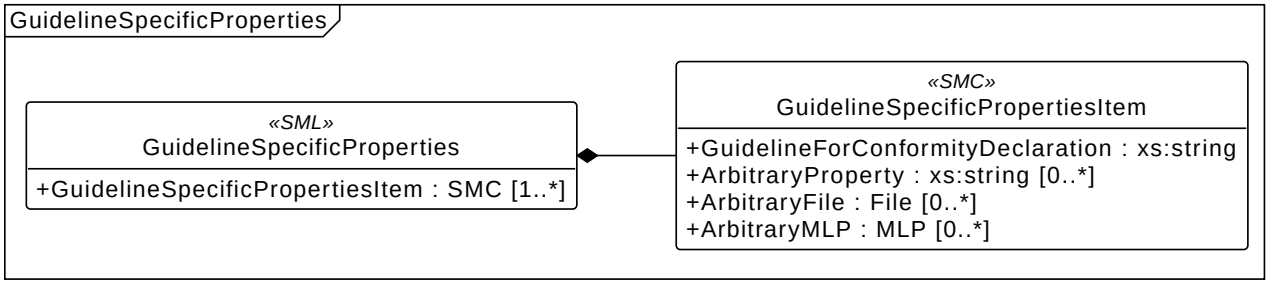


Figure 21. UML for guideline specific properties

For a single set of guideline specific properties, the SubmodelElements are described as follows. The table convention is explained in Annex A.2.

idShort:	**GuidelineSpecificProperties**		
Class:	SubmodelElementList		
semanticId:	https://admin-shell.io/idta/cds/GuidelineSpecificProperties/1		
Parent:	AssetSpecificProperties		
Explanation:			
Element details:	typeValueListElement=SubmodelElementCollection		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[SMC] GuidelineSpecificPropertiesItem	0173-1#01-AHD205#004	[] 4 elements	1..*

idShort:	**GuidelineSpecificPropertiesItem**
Class:	SubmodelElementCollection
semanticId:	0173-1#01-AHD205#004

Parent:	GuidelineSpecificProperties		
Explanation:			
Element details:	-		
[SME type] idShort	semanticId Description@en	[valueType] example	card.
[Prop] GuidelineForConformityDeclaration	0173-1#02-AAO856#002 Reference to a guideline or document used to demonstrate conformity of the described connection.	[String]	1
[Prop] ArbitraryProperty	https://admin-shell.io/SMT/General/ArbitraryProp Note: Every property can be used. Note: The idShort is arbitrary Note: The use of a displayName is recommended.	[String]	0..*
[File] ArbitraryFile	https://admin-shell.io/SMT/General/ArbitraryFile Note: Every file can be used. The idShort is arbitrary Note: The use of a displayName is recommended.	[]	0..*
[MLP] ArbitraryMLP	https://admin-shell.io/SMT/General/ArbitraryMLP Note: Every multilingual property can be used. Note: The idShort is arbitrary Note: The use of a displayName is recommended.	[] ""sample""@en	0..*

Annex A. Explanations on used table formats

1. General

The used tables in this document try to outline information as concise as possible. They do not convey all information on Submodels and SubmodelElements. For this purpose, the definitive definitions are given by a separate file in form of an AASX file of the Submodel template and its elements.

2. Tables on Submodels and SubmodelElements

For clarity and brevity, a set of rules is used for the tables for describing Submodels and SubmodelElements.

- The tables follow in principle the same conventions as in [5].
- The table heads abbreviate 'cardinality' with 'card'.
- The tables often place two informations in different rows of the same table cell. In this case, the first information is marked out by sharp brackets [] form the second information. A special case are the semanticIds, which are marked out by the format: (type)(local)[idType]value.
- The types of SubmodelElements are abbreviated (see Table 24):

Table 24. Abbreviations for SubmodelElements

SME type	SubmodelElement type
Blob	Blob
Cap	Capability
Ent	Entity
Evt	Event
File	File
MLP	MultiLanguageProperty
Opr	Operation
Prop	Property
Range	Range
Ref	ReferenceElement
Rel	RelationshipElement
RelA	AnnotatedRelationshipElement
SMC	SubmodelElementCollection
SME	SubmodelElement type
SML	SubmodelElementList

- If an idShort ends with '__00__', this indicates a suffix of the respective length (here: 2) of decimal digits, in order to make the idShort unique. A different idShort might be choosen, as long as it is unique in the parent's context.
- The Keys of semanticId in the main section feature only idType and value, such as: https://admin-shell.io/vdi/2770/1/0/DocumentId/Id. The attribute "type" (typically "ConceptDescription" and "(local)" or

"GlobalReference") need to be set accordingly; see [6].

- If a table does not contain a column with "parent" heading, all represented attributes share the same parent. This parent is denoted in the head of the table.
- Multi-language strings are represented by the text value, followed by '@'-character and the ISO 639 language code: example@EN.
- The [valueType] is only given for Properties.

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